→ SVM - 2

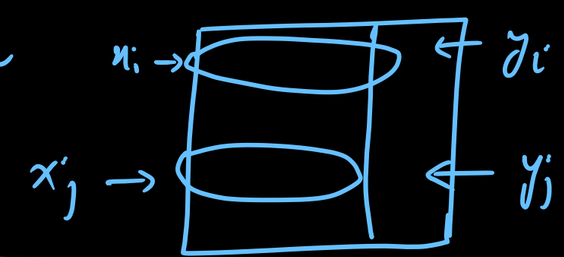
Soft Margin

$$L = \min_{w, b, \xi_i} \left(\|w\|^2 + C \sum \xi_i \right)$$

$$\xi_i \geq 0$$

$$\text{s.t} \quad y_i \left(w^T x_i + b \right) \geq 1 - \xi_i$$

→ Primal form



Dual Form

$$L = \max \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j x_i^T \cdot x_j$$

$$\text{s.t} \quad 0 \leq \alpha_i \leq C$$

$$\sum \alpha_i y_i = 0$$

→ Langragian Method

→ $\frac{\partial L}{\partial w} = 0 \quad \frac{\partial L}{\partial b} = 0$

→ KKT condition

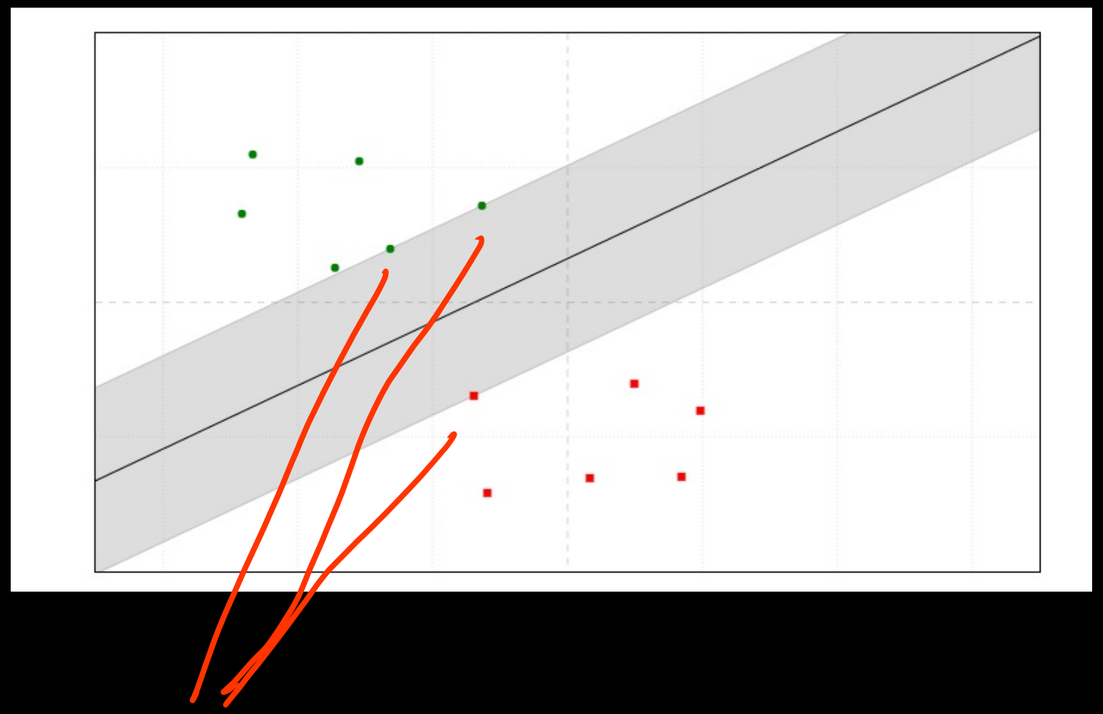
$$\begin{cases} L = \min\limits_{w,b,\xi_i} \left(\|w\|^2 + C \sum \xi_i \right) \\ \text{s.t.} \quad ① \ \xi_i \geq 0 \\ \qquad ② \ y_i \left(w^T x_i + b \right) \geq 1 - \xi_i \end{cases}$$

$$L = \min_{w,b,\xi_i} \left(w^T w + C \sum \xi_i \right) + \sum \alpha_i \left(y_i (w^T x_i + b) - 1 + \xi_i \right) + \sum \beta_i \xi_i$$

→ Support Vectors

↳ Point which lies on margin

↳ Only support vector affect decision boundary.



Support Vector

x_q

$$\hat{y} = \text{sign}(w^T x_q + b)$$

$$\text{sign}(x) = 1 \quad x \geq 0$$
$$= 0 \quad x < 0$$

$$\boxed{w = \sum_{i=1}^{N} \alpha_i y_i x_i^T}$$

$\alpha_i = 0$

$c > \alpha_i > 0 \Rightarrow$ Support vector

$$\hat{y} = \text{sign}\left(\sum_{i=1}^{N} \alpha_i y_i x_i^T \cdot x_q + b \right)$$

$$= \text{sign}\left(\sum_{x_i \in SV} \alpha_i y_i x_i^T x_q + b \right)$$

→ 10,000 training data point

1% SV → 100 data point

$$\hat{y}_q = \text{sign}\Big(\sum_{x_i \in SV} \alpha_i y_i x_i^T x_q + b \Big)$$

$\alpha_i = 0$

$C > \alpha_i > 0$ SV

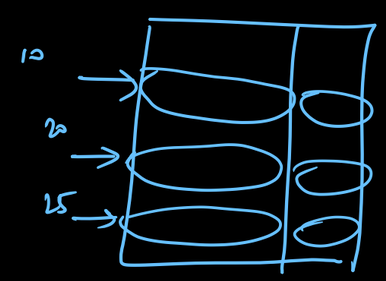
few training data point which were support vector

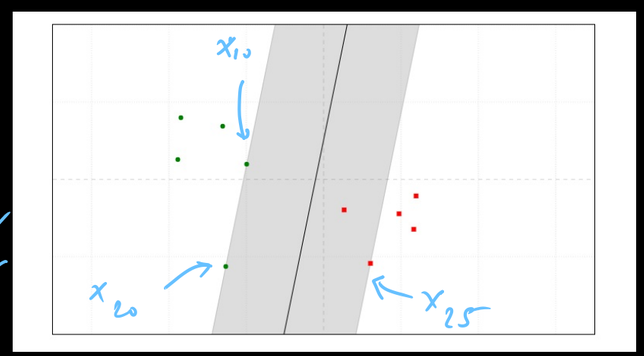
→ 300 data point

1% SV → 3

$$\hat{y}_q = \text{sign}\Big(\sum_{x_i \in SV}^{N} \alpha_i y_i x_i^T x_q \Big)$$

$$= \alpha_{10} y_{10} x_{10}^T x_q + \alpha_{20} y_{20} x_{20}^T x_q + \alpha_{25} y_{25} x_{25}^T x_q$$





$$\boxed{0 < \alpha < C}$$

Summary

SV

1) α_i for SV

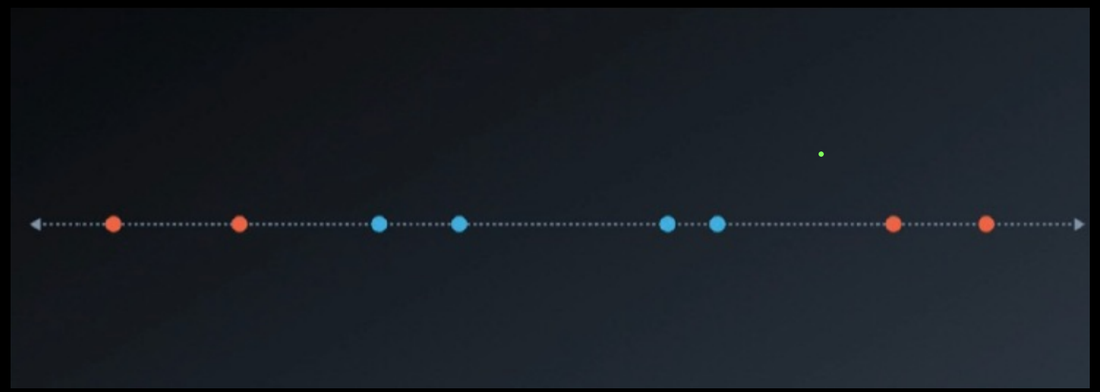
2) x_i, y_i for SV

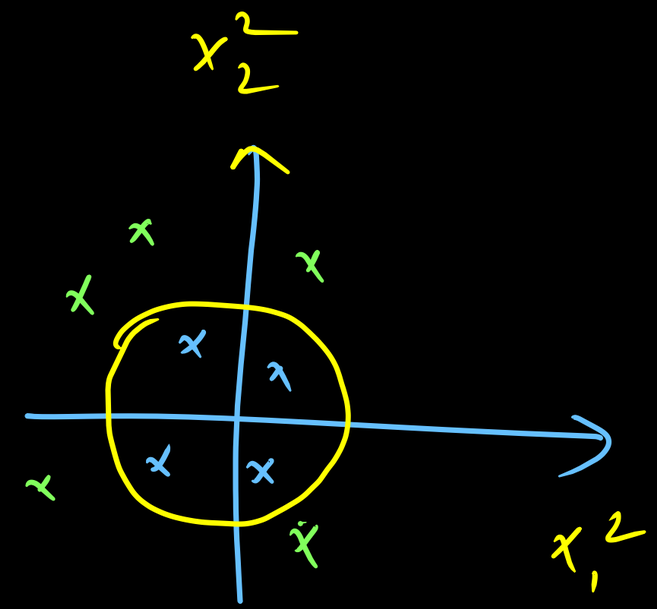
Hard Margin

Soft Margin

↳ Can only seperate linear data

→ Kernel SVM → Non-linear data

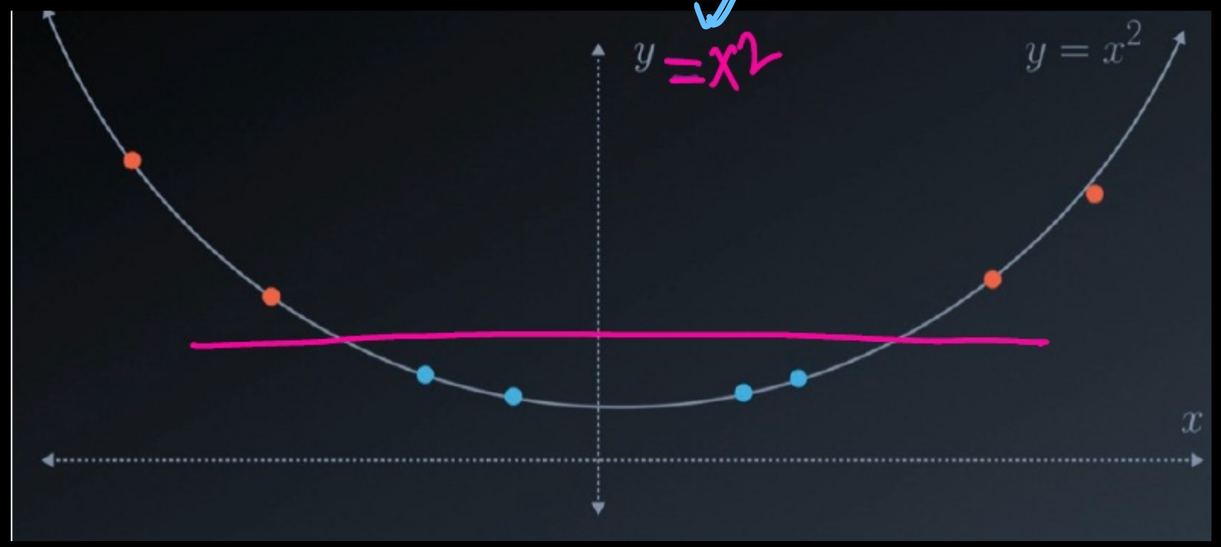


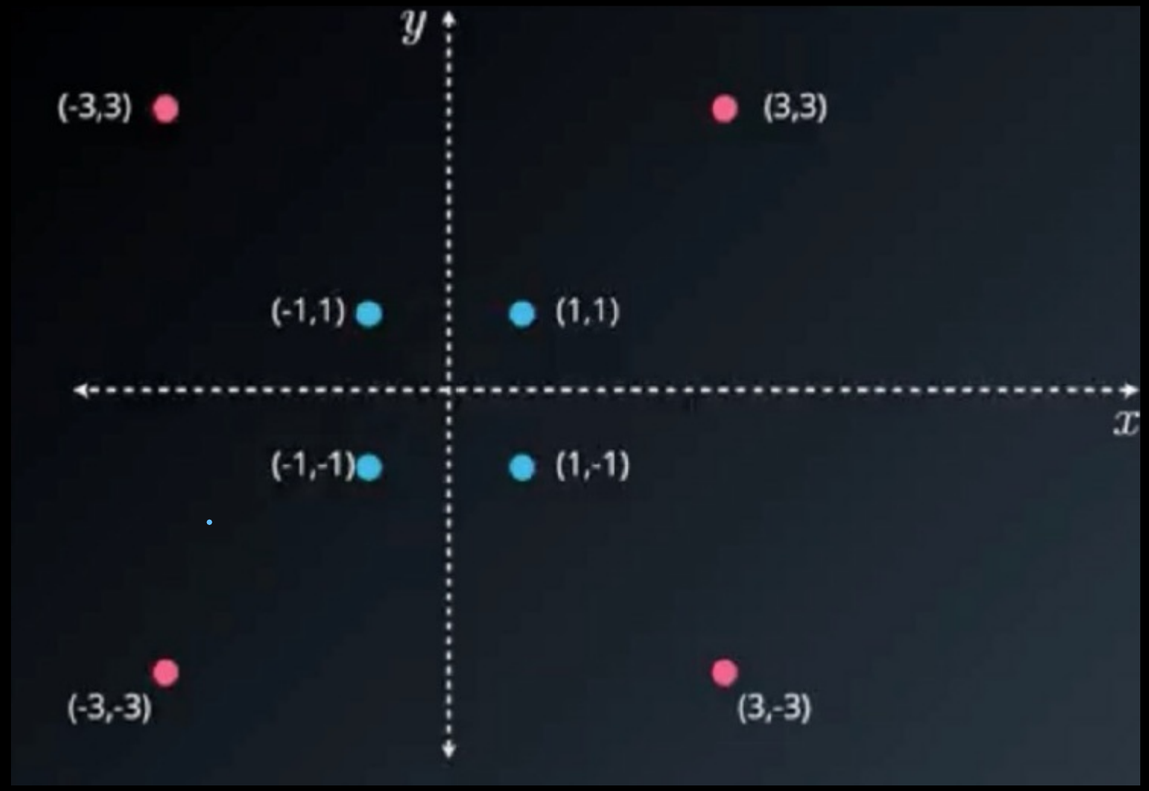


$$Z = w^T x + b$$

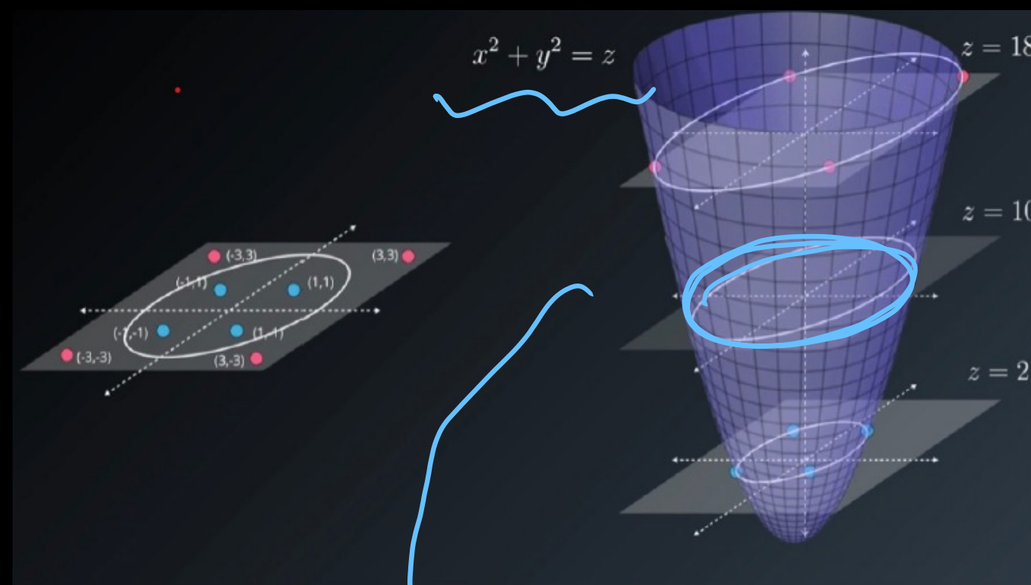
$$= w_1 x_1 + w_{12} x_1^2 + w_2 x_2 + w_{22} x_2^2$$

$$w_1 = 0, \quad w_2 = 0$$

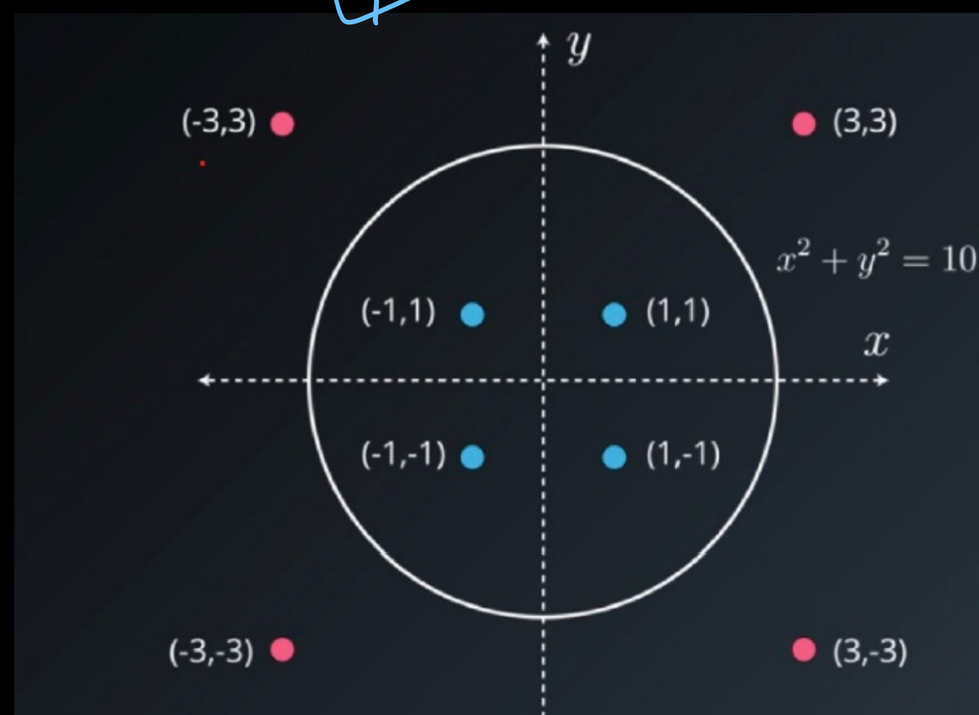

$y = x^2$
$y = x^2$
x


y
x
(-3,3)
(3,3)
(-1,1)
(1,1)
(-1,-1)
(1,-1)
(-3,-3)
(3,-3)

$$z = x^2 + y^2$$


$x^2 + y^2 = z$
$z = 18$
$z = 10$
$z = 2$
(-3,3)
(3,3)
(-1,1)
(1,1)
(-1,-1)
(1,-1)
(-3,-3)
(3,-3)

$= x^2 + y^2$


y
x
$x^2 + y^2 = 10$
(-3,3)
(3,3)
(-1,1)
(1,1)
(-1,-1)
(1,-1)
(-3,-3)
(3,-3)

2D $\longrightarrow$ 10D

x^2, x^3, x^4

Dual Form

$$\boxed{\begin{aligned} L = \max \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \phi(x_i) \cdot \phi(x_j) \\ \text{s.t} \quad 0 \le \alpha_i \le C \\ \sum \alpha_i y_i = 0 \end{aligned}}$$

$$\underbrace{\phi(x_i) \cdot \phi(x_j)}_{\text{Compute Intensive}} = \underbrace{K(x_i, x_j)}_{\text{Not compute Intensive}}$$

2D $\longrightarrow$ 10D

→ Break until 22:24 PM

$$K(x_1, x_2) = \underbrace{(1 + x_1^T \cdot x_2)^2}_{2D}$$

$$= \left(1 + \begin{bmatrix} x_{11} & x_{12} \end{bmatrix} \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}\right)^2$$

$$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix}$$

$$x_2 = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}$$

$$= (1 + x_{11} x_{21} + x_{12} x_{22})^2$$

$$= 1 + x_{11}^2 x_{21}^2 + x_{12}^2 x_{22}^2 + 2 x_{11} x_{21} + 2 x_{12} x_{22}$$

$$+ 2 x_{11} x_{21} x_{12} x_{22}$$

$$\phi(x_1) = \overbrace{\left[1,\ x_{11}^2\ \ x_{12}^2,\ \sqrt{2}\,x_{11}^2,\ \sqrt{2}\,x_{12}^2,\ \sqrt{2}\,x_{11}x_{12}\right]}^{6D}$$

$$\phi(x_2) = \left[1,\ x_{21}^2,\ x_{22}^2,\ \sqrt{2}\,x_{21}^2,\ \sqrt{2}\,x_{22}^2,\ \sqrt{2}\,x_{21}\,x_{22}\right]$$

$$\phi(x_1)\cdot\phi(x_2)$$

6D

$$= 1 + x_{11}^2 x_{21}^2 + x_{12}^2 x_{22}^2 + 2x_{11}^2 x_{21}^2 + 2x_{12}^2 x_{22}^2$$

$$+ 2\,x_{11}x_{12}x_{21}x_{22}$$

$$K(x_1, x_2) = \underbrace{(1 + x_1^T x_2)}_{2D}{}^2 = \underbrace{\phi(x_1)\cdot\phi(x_2)}_{6D}$$

→ linear

$$K(x, x') = x^T x'$$

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

→ Polynomial

$$K(x, x') = (x^T x' + c)^d$$

→ RBF (Radial Basis Function)

$$K(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right)$$

$$= \exp\left(-\frac{d^2}{2\sigma^2}\right) \qquad d = \|x - x'\|$$

$$p = \frac{d^2}{2\sigma^2}$$

$$= 1 + p + \frac{p^2}{2!} + \frac{p^3}{3!} - - \frac{p^K}{K!}$$

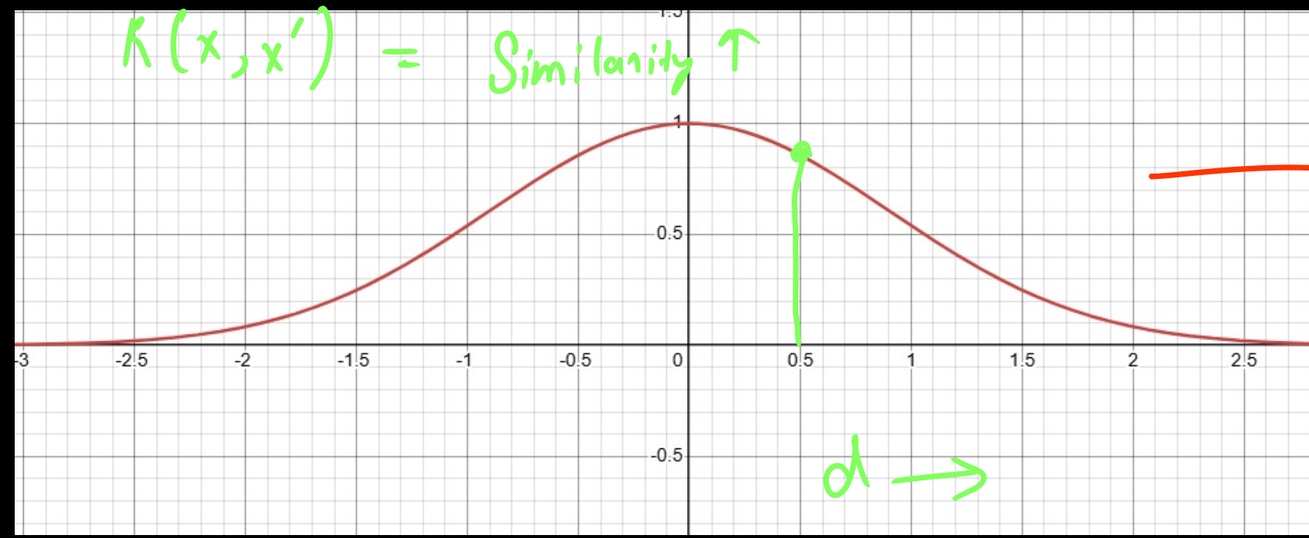
$\boxed{\infty\text{-dimension}}$

$$K(x, x') = \exp\left(-\frac{d^2}{2\sigma^2}\right)$$

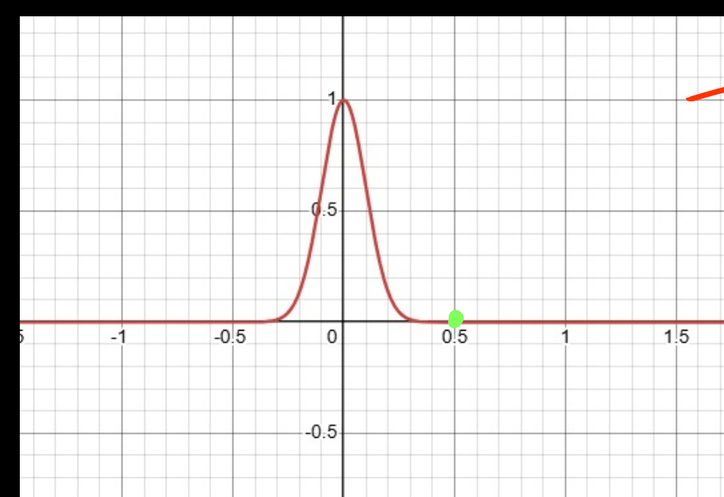
$$d = \|x - x'\|$$

$d \downarrow \rightarrow$ Similarity $\uparrow$

$d \uparrow \rightarrow$ Similarity $\downarrow$



$\sigma = 0.8$



$\sigma = 0.1$

Time Complexity

Test time

$$\hat{y} = \text{sign}\left(\sum_{x_i \in SV} \alpha_i y_i \overset{\nearrow d}{x_i^T x_q} + b \right)$$

$\hookrightarrow O(\text{no of SV} * d)$

KNN $\rightarrow O(md)$

Training

→ $O(n^3)$

↳ $O(n^2)$

↳ Time Complexity

↳ SVM is slow for training.

SVM